

COMP2300 Set 5 Practice Exam Solutions

Part A: Multiple Choice

- A1. B
- A2. C
- A3. B
- A4. D
- A5. C
- A6. B
- A7. B
- A8. A
- A9. B
- A10. B

Part B: Computational Problems

B1. State sequences and cycle counts

- (a)
 - LDR: Fetch -> Decode -> MemAdr -> MemRead -> MemWB, so 5 cycles.
 - STR: Fetch -> Decode -> MemAdr -> MemWrite, so 4 cycles.
 - DP immediate: Fetch -> Decode -> ExecuteI -> ALUWB, so 4 cycles.
 - B: Fetch -> Decode -> Branch, so 3 cycles.
- (b) The 6-instruction sequence takes:

$$5 + 4 + 4 + 3 + 4 + 5 = 25 \text{ cycles}$$

Average CPI:

$$\frac{25}{6} \approx 4.17$$

B2. PC, R15, and branch target calculation

- (a) Reading R15 should supply PC + 8, so:

$$0x80A0 + 8 = 0x80A8$$

(b)

$$\text{ExtImm} = \text{sign-extend}(0x000003 \ll 2) = 0x0000000C$$

$$\text{BTA} = \text{PC} + 8 + \text{ExtImm} = 0x80A8 + 0x0000000C = 0x80B4$$

B3. Decoder outputs

Instruction class	Op	RegSrc1:0	ImmSrc
LDR	01	10	01
STR	01	10	01
DP immediate	00	00	00
DP register	00	00	00
B	10	01	10

Explanation:

- $\text{RegSrc0} = 1$ only when $\text{Op} = 10$.
- $\text{RegSrc1} = 1$ only when $\text{Op} = 01$.
- $\text{ImmSrc} = \text{Op}$.

B4. Average CPI and total cycles

(a)

$$\begin{aligned}\text{Average CPI} &= (0.25)(5) + (0.10)(4) + (0.13)(3) + (0.52)(4) \\ &= 1.25 + 0.40 + 0.39 + 2.08 = 4.12\end{aligned}$$

(b) Total cycles for 50 million instructions:

$$50 \times 10^6 \times 4.12 = 206 \times 10^6$$

So the total is 206 million cycles.

B5. Critical path and clock frequency

(a) First compute:

$$t_{ALU} + t_{mux} = 120 + 25 = 145 \text{ ps}$$

Hence:

$$\max(145, 200) = 200 \text{ ps}$$

Therefore:

$$T_{c2} = 40 + 2(25) + 200 + 50 = 340 \text{ ps}$$

(b)

$$f_{max} = \frac{1}{340 \times 10^{-12}} \approx 2.94 \times 10^9 \text{ Hz}$$

So:

$$f_{max} \approx 2.94 \text{ GHz}$$

B6. Execution time

(a)

$$\begin{aligned} T &= N \times CPI \times T_c \\ &= (100 \times 10^9)(4.12)(340 \times 10^{-12}) \\ &= 140.08 \text{ s} \approx 140 \text{ s} \end{aligned}$$

(b) Revised design:

$$T' = (100 \times 10^9)(4.12)(300 \times 10^{-12}) = 123.6 \text{ s}$$

(c) Speedup:

$$\text{Speedup} = \frac{140.08}{123.6} \approx 1.13$$

Part C: Past Exam Questions

C1. ISA vs Microarchitecture

- (a) Ripple-carry vs carry-save adder: **Microarchitecture**
- (b) Instruction width: **ISA**
- (c) Branch prediction: **Microarchitecture**
- (d) Hardwired vs microprogrammed control: **Microarchitecture**

C2. Multi-Cycle Execution Time

Average CPI:

$$(0.50)(6) + (0.25)(8) + (0.25)(4) = 3 + 2 + 1 = 6$$

Execution time:

$$T = \frac{2 \times 10^9 \times 6}{1 \times 10^9} = 12 \text{ s}$$

C3. Single-Cycle vs Pipelining

Two drawbacks of a single-cycle processor are:

- The clock period must be long enough for the slowest instruction.
- Hardware must be provisioned for worst-case resource use, so utilization is poor.

Pipelining addresses these by overlapping different stages of different instructions, increasing throughput and making better use of otherwise idle hardware.

ILP is the ability to execute or make progress on multiple independent instructions at the same time. Two microarchitecture types that exploit ILP are **superscalar** and **VLIW**.

C4. ILP Ranking

Highest to lowest ILP potential:

$$A > B > C$$

where:

- A = scalar in-order pipeline with forwarding and hazard detection
- B = scalar in-order pipeline without forwarding or hazard detection
- C = multi-cycle processor

Justification:

- A can overlap instructions while also reducing stalls via forwarding and hazard handling.
- B still overlaps instructions, but extra NOPs and stalls reduce usable ILP.
- C has the least ILP because it does not pipeline instruction stages across multiple instructions.